ICML 2026 · POSTER

Progressive Cramming: Reliable Token Compression and What It Reveals

One embedding reconstructs >1000 tokens — yet perfect reconstruction does not preserve useful semantics.

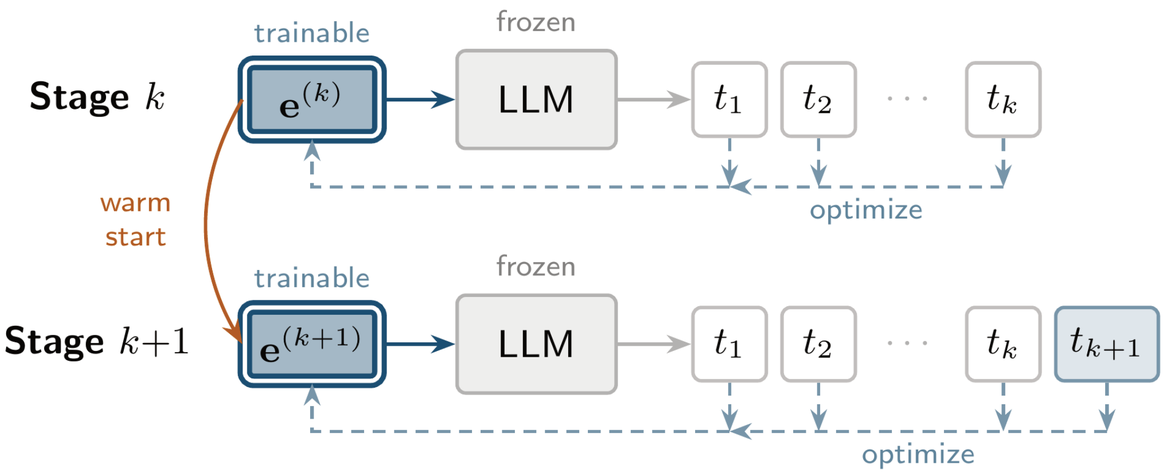
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❶ Progressive Cramming

Train one embedding for a **frozen LM**. Grow the target **one token at a time**; advance only after perfect reconstruction.



→ **100% greedy reconstruction** at every accepted stage.

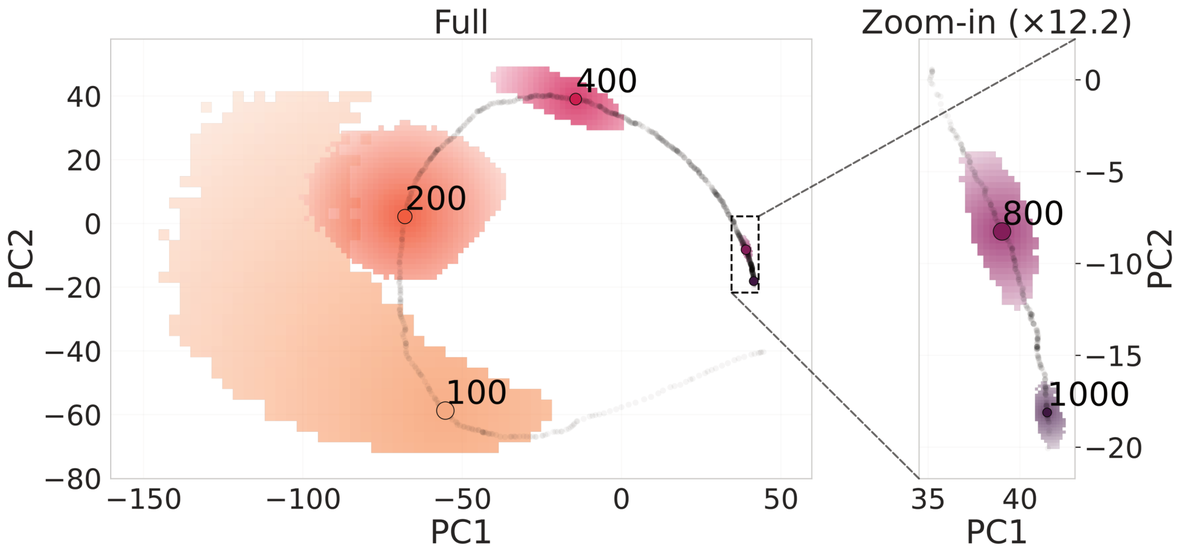
→ Stop at first failure; save the embedding trajectory.

❷ Optimisation Trajectories

30–100	**~log**
PCA components for 99% variance	growth with sequence length

→ **Low-dimensional paths:** a few PCA components explain 99% of variance.

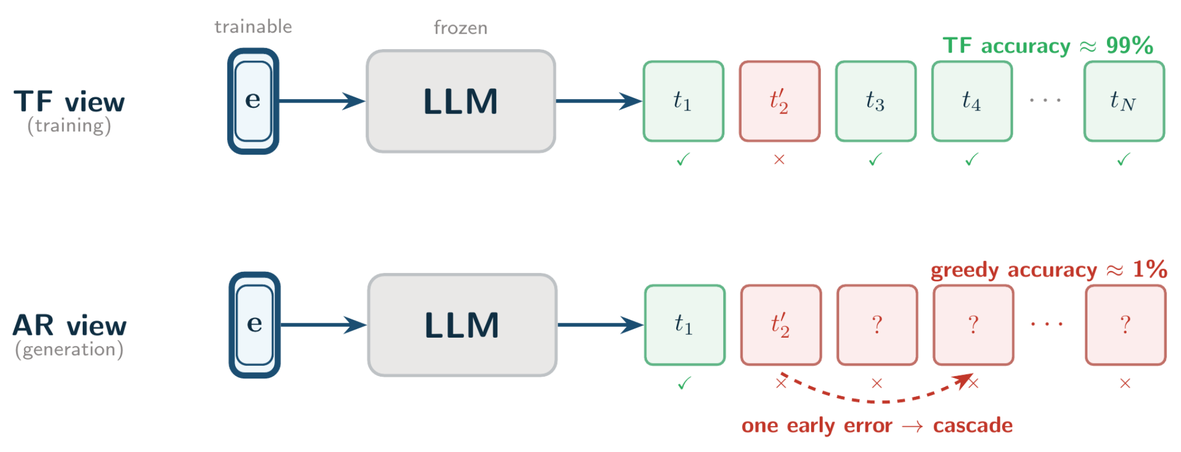
→ Longer prefixes leave **smaller accuracy basins**.



Llama-3.1-8B. PC1–PC2 explain 65.7% of trajectory variance. Color: reconstruction accuracy.

❸ Why Not Full Cramming?

Full cramming fixes length; progressive cramming fixes **100% reconstruction**. Small teacher-forcing errors can cascade during generation:



Schematic: early errors can derail generation.

Model / Type	Tokens	TF Acc	Greedy
Llama-8B / Full	1568	99.96%	40.4%
Llama-8B / Progr.	**1438**	**100%**	**100%**
Pythia / Full	512	99.71%	44.2%
Pythia / Progr.	**430**	**100%**	**100%**

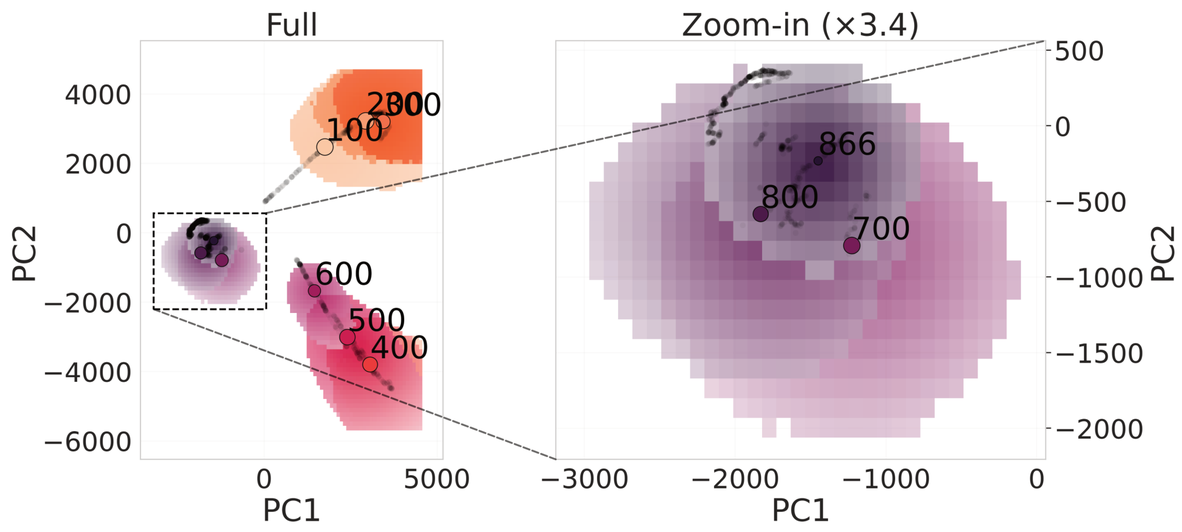
❹ Across Model Families

Mean compressed tokens on **50 PG19 samples**:

Model	Tokens	+LowDim	Boost
Llama-3.1-8B	1438	**1697**	1.2×
Pythia-1.4b	430	500	1.2×
SmolLM2-1.7B	335	957	2.9×
Gemma-3-4B	214	697	**3.3×**

Up to 3.3× more tokens.

Low-dim projection: jointly optimize per-sample W, z, b; $e = Wz + b$, with $z \in \mathbb{R}^k$, $k \ll d$.



With low-dim projection. PC1–PC2: 61.0% variance. Color: >90% reconstruction regions.

❺ Reconstruction ≠ Understanding

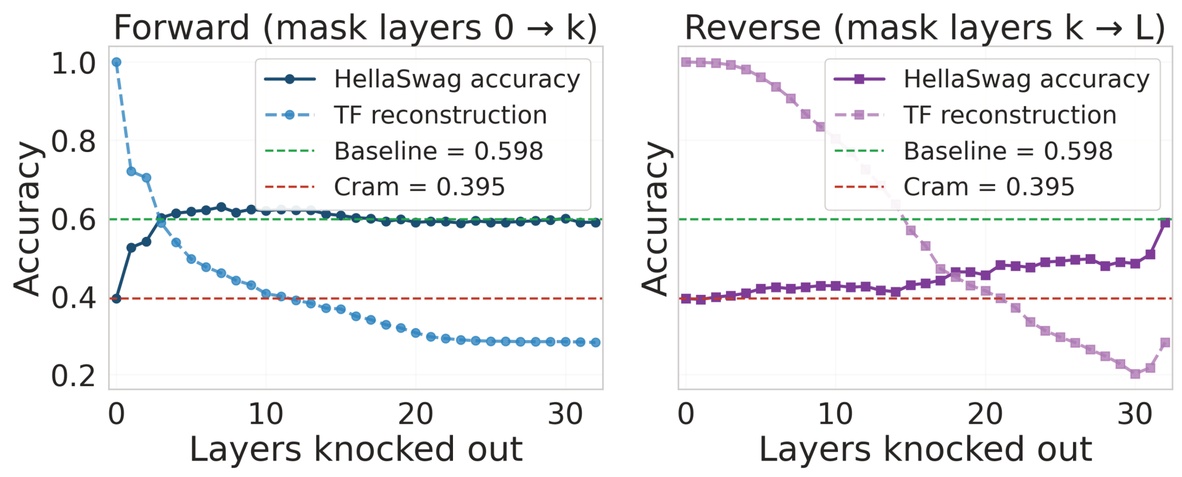
Prepending the optimized embedding **reduces accuracy**, even with the original prefix in context:

Model	HS Base	HS +Emb	ARC Base	ARC +Emb
Pythia-1.4b	44.0%	37.6%	53.1%	49.1%
SmolLM2-1.7B	53.6%	37.0%	67.3%	56.0%
Llama-3.1-8B	61.9%	40.8%	63.5%	43.3%

HS = HellaSwag; ARC = ARC-E. +Emb = original context with the optimized embedding.

Generative 5-shot MMLU: accuracy falls to **~0%**. A random embedding stays near baseline.

❻ Attention Knockout



Llama-3.1-8B. Masking early layers restores downstream accuracy; masking late layers does not.

Early layers drive the disruption. The same pattern holds for Pythia-1.4B and SmolLM2-1.7B.

Takeaways & Next Steps

→ **Reconstruction ≠ semantics.** Perfect decoding can coexist with downstream collapse.

→ **Optimization headroom.** Low-dim projection improves capacity; useful semantic compression remains open.